Unit 2 Readiness Check Teacher Key

3 items · 3 points each · 9 total. Award 1 for a valid model/setup, 1 for correct mathematics/units, and 1 for a requested justification. Use the applicable evidence when the task is conceptual; accept equivalent methods and response modes. Do not count writing length as mathematical mastery.

Item 1

Prompt: A learner records the length of every pencil in a box. Is asking about these lengths statistical? Name the population, variable, and unit.

Solution / evidence: Yes; anticipated variability across lengths. Population: pencils in the box. Variable: length. Unit: centimeters or another appropriate length unit.

Item 2

Prompt: Find the median of 8, 2, 6, 4. Show the ordering and explain the two middle positions.

Solution / evidence: Order 2, 4, 6, 8; median 5.

Item 3

Prompt: Compare IQR for [2, 4, 6, 8] and [2, 4, 6, 28]. Use medians of the two halves. Explain whether changing the maximum can affect IQR.

Solution / evidence: First IQR 4; second IQR 14. It can affect Q3 in this four-value set.

Response: group errors by interpretation/model versus calculation versus explanation. Use the corresponding lesson Teaching Support Guide, revise one step, and reassess with its reserved fresh check. Do not reuse the already seen form as independent evidence.